

Medical Unmanned Aerial Vehicle System: Command and Control Center

Bayu Dimas Febriansyah

The healthcare system often needs fast and safe transport of various goods between facilities to function. This includes items such as blood, organs, and various types of medical supplies that need to reach hospitals, urgent cares, and research institutes in an efficient manner. To this end, the Medical Unmanned Aerial Vehicle (MUAV) system aims to compete on cost, while offering quicker, on-demand delivery of perishable and/or niche medical cargo.

This report will briefly present an overview of the entire system-of-systems (SOS) involved, before delving into the specifics of the Command and Control Center (CCC). The CCC serves as the operational hub of the Medical UAV System. It manages mission planning, fleet supervision, and real-time communication with all delivery vehicles. The CCC receives transport requests, generates flight plans, coordinates with air traffic services, monitors telemetry and payload status during flight, and executes rerouting or safety interventions when necessary. It also interfaces with ground stations, customer systems, maintenance services, and external navigation sources such as GPS. In essence, the CCC orchestrates and synchronizes every component of the MUAV network, ensuring safe, efficient, and timely medical deliveries.

I. System of Systems Overview

A. History of the Problem

For centuries, moving the sick and wounded was slow and painful, relying on human power via litters and basic animal-drawn carts. The primary challenge was the lack of speed and dedicated infrastructure, meaning transport often occurred after the conflict or crisis had passed. A significant shift occurred during the Napoleonic Wars when Dominique Jean Larrey pioneered the concept of the Flying Ambulance, which are lightweight, horse-drawn carriages designed to rapidly evacuate the wounded during battle. This innovation established timeliness as a critical factor in patient outcome. The 19th century saw further systematization, especially during the American Civil War, and the introduction of dedicated civilian carriages for managing public health crises like cholera outbreaks.

The true transformation came with mechanization. The introduction of the motorized ambulance around 1900 provided speed and reliability unmatched by horses, fundamentally altering urban emergency response. This was swiftly followed by the use of fixed-wing aircraft and, critically, helicopters during the World Wars and the Korean conflict. Helicopters proved vital because they could bypass all ground obstacles, offering fast, point-to-point evacuation from remote or difficult terrain. In the late 20th century, the focus shifted to in-transit care, with ambulances evolving into mobile intensive care units, while refrigerated trucks and dedicated couriers became essential for maintaining the integrity and speed of the medical supply chain.

B. Mission Background and Objective

Medical supplies, blood products, and donor organs are highly time sensitive items that must be couriered quickly and efficiently to provide life-saving care to patients. Currently, these deliveries are conducted by either land-based transportation or helicopters, the former of which are relatively slow, and the latter of which carry a fairly high cost.

This also leads to the issue of personnel availability, as only authorized couriers are allowed to transport blood products and organs due to both patient safety and regulatory concerns. As such, a system which uses a fleet of Unmanned Aerial Vehicles (UAVs) to transport these fragile and time sensitive products would bolster the ability of the healthcare system to save lives.

The mission's main objective is to deploy an autonomous on-demand aerial logistics network capable of rapidly transporting volatile or niche cargo to local healthcare facilities in order to improve patient standard of care reducing ground-transportation costs.

C. Mission Overview

The primary mission of the Medical Unmanned Aerial Vehicle (MUAV) is to establish an on-demand, autonomous aerial logistics system capable of bypassing ground-based transportation inefficiencies. This System of Systems (SoS) addresses the critical need for the rapid transport of time-sensitive biological assets, such as blood products, donor organs, and pharmaceuticals, which are currently constrained by urban traffic congestion and the high operational costs of manned rotary-wing aviation. By utilizing a fleet of autonomous vehicles equipped with specialized environmental payload modules and paired with command center, this SoS ensures the secure and viable delivery of volatile cargo to hospitals and urgent care facilities. This will ultimately improve the standard of care and lowering costs for the patient.

D. System Description

1. System Characteristics

- The Medical Unmanned Aerial Vehicle (MUAV) is an integrated aerial logistics network composed of coordinated drones, payload interfaces, and a centralized Command and Control Center (CCC).
- The system is autonomous by design, relying on mission automation and Beyond Visual Line of Sight (BVLOS) operations rather than direct piloting to achieve rapid, scalable transport.
- The system is mission-oriented toward medical logistics, specifically tailored to transport volatile, niche, or time-critical healthcare materials between distributed facilities.
- The system is distributed and multi-node, with missions initiated from various healthcare facilities and executed across multiple drones and routes simultaneously.
- The system is environment-adaptive, functioning across varied weather, airspace conditions, and ground infrastructures, including constrained delivery zones.
- The system is safety-critical, requiring high reliability, continuous monitoring, and coordinated failsafes to protect cargo, users, and the public.
- The system is data-centric, dependent on real-time telemetry, airspace information, and payload condition data to guide operational decisions.

2. System Attributes

- Uses specialized aerial Delivery Vehicles in the form of UAVs to physically carry the medical payload.
- Uses a Command and Control Center to efficiently coordinate multiple drones without direct manual piloting.
- Uses Payload Delivery Modules to secure and deliver medical cargo throughout the mission lifecycle.
- Uses real-time communication links between the Delivery Vehicles, Payload Delivery Modules, and the Command and Control Center to maintain mission oversight and compliance.
- Uses scalable architectures that allow the system to expand its fleet and mission volume without extensive redesign.

3. *System Capabilities*

1) **The ability to execute on-demand deployment**

- It is the ability to launch aerial assets immediately upon receiving a validated request from a healthcare facilities. It is needed to minimize transportation time compared to ground services. This addresses the critical time-sensitivity of medical supplies.

2) **The ability to navigate in complex urban airspace**

- The system must autonomously avoid buildings, towers, and other aircraft. Reliable obstacle avoidance is necessary for operating safely over populated regions and for meeting FAA airspace requirements.

3) **The ability to coordinate fleet logistics**

- This involves managing multiple UAVs at once. It involves deciding which asset to dispatch, how to route around weather, and how to balance battery levels and demand. Effective coordination prevents bottlenecks when requests spike and helps keep the network running efficiently.

4) **The ability to maintain continuous communication**

- The system needs a stable command-and-control link across the operating region. This continuous communication enables real-time updates, telemetry monitoring, and remote pilot intervention if something unexpected occurs.

5) **The ability to monitor environmental conditions**

- This includes gathering localized weather data such as wind shear, precipitation, and visibility. Having that information allows the system to reroute or delay flights when conditions exceed design limits. This reduces the risk of losing an aircraft or damaging the payload.

6) **The ability to manage system health and maintenance**

- Fleet-wide tracking of battery condition, usage cycles, and component wear is required. This capability supports predictive maintenance and helps ensure that any UAV with a potential fault is identified before it is assigned to a mission.

7) **The ability to interface with national air traffic services**

- The system must be able to exchange flight plans and transponder data with ATC and unmanned service suppliers. This is due the fact that the medical UAVs must safely operate alongside helicopters and commercial aircraft in shared airspace.

8) **The ability to process secure medical orders**

- This includes receiving, validating, and prioritizing digital requests from hospital systems. It ensures the correct medical item is assigned to the correct mission while meeting all handling and regulatory requirements for sensitive biological materials.

E. Concept of Operations

The Medical UAV System operates as an integrated logistics network that connects healthcare facilities with autonomous aerial delivery services. The process begins when a medical facility submits a transport request through the Customer Interface, which forwards mission details to the Command and Control Center (CCC). The CCC evaluates requirements, checks aircraft availability, and generates a flight plan.

Once authorized, the CCC assigns a Delivery Vehicle and instructs the Ground Station to prepare and deploy it. The cargo is loaded into the Payload Delivery Module, which verifies proper sealing and environmental conditions before attaching to the aircraft. The Delivery Vehicle receives navigational data from GPS and airspace clearances from Air Traffic Services, enabling safe BVLOS flight. During flight, the Delivery Vehicle continuously reports telemetry, payload status, and system health to the CCC. The CCC monitors progress, updates routing if conditions change, and

coordinates with Maintenance Services if anomalies arise.

Upon arriving at the destination, the Delivery Vehicle performs an autonomous landing or precise drop, and the Payload Delivery Module authenticates the delivery location before releasing the cargo. The Customer Interface notifies the medical facility of delivery completion. The aircraft then returns to a Ground Station or proceeds to its next assignment, while Maintenance Services update readiness status for future missions. This operational model allows the Medical UAV System to provide fast, reliable, and automated medical logistics, with each subsystem coordinating seamlessly through the CCC as shown in the diagram.

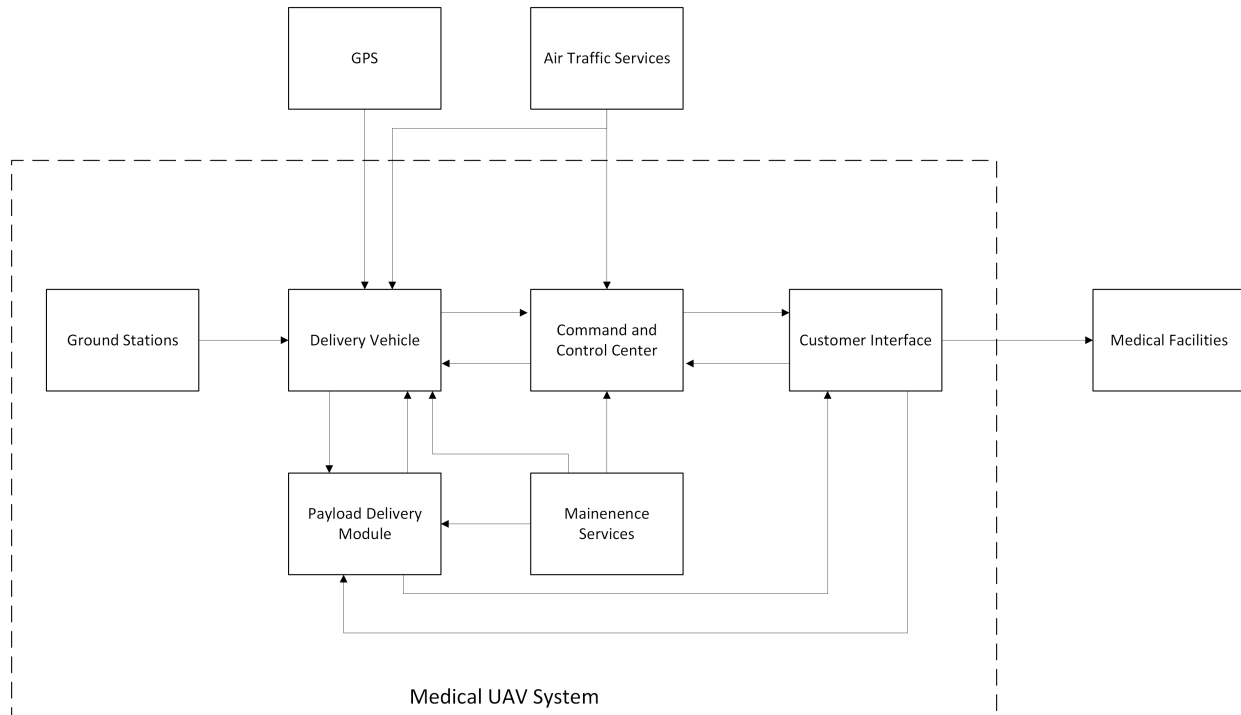


Fig. 1 Medical Unmanned Aerial Vehicle System of Systems OV-1

II. System of Interest Overview

A. History

The Command and Control Center (CCC) has evolved from manual military war rooms to today's autonomous fleet-management hubs. Early CCCs (1940s–1970s), such as the Dowding System, and later SAGE and NASA Mission Control, marked the shift from analog plotting tables to digital mainframes capable of centralized data processing and continuous telemetry. From the 1980s to 2010s, advances in networking transformed CCCs into real-time, distributed systems. Air Traffic Control centers and Network Operations Centers introduced automated alerting, geospatial mapping, and large-scale monitoring of distributed assets. The rise of unmanned and autonomous vehicles in the 2010s to today pushed CCCs into their modern form: operator-supervised, software-driven control hubs capable of managing entire fleets. Medical Unmanned Aerial Vehicle (MUAV) CCCs rely on automation, anomaly detection, and Unmanned Traffic Management (UTM)-based airspace integration, enabling fast, safe, and coordinated medical logistics.

B. Objective

The Command and Control Center (CCC) is a System of Interest (SoI) within the Medical Unmanned Aerial Vehicle (MUAV) System-of-Systems, responsible for orchestrating mission planning, fleet coordination, safety oversight, and communications across all MUAV operations. The objective of the Communications and Command Center (CCC) is to provide continuous, safe, and reliable command-and-control, communication, and mission management capabilities for all Medical Unmanned Aerial Vehicle (MUAV) operations. It ensures that every MUAV mission is planned, authorized, monitored, communicated, and executed within regulatory, airspace, safety, and medical logistics constraints.

C. Characteristics

- 1) Physically housed in a secured operations facility that contains mission-control workstations, communication servers, visualization screens, and UTM/ATC interface equipment.
- 2) Incorporates multi-layer communication infrastructure including cellular (4G/5G), RF line-of-sight links, satellite channels, and encrypted Wi-Fi for ground operations.
- 3) Integrates several subsystems: mission management suite, telemetry and communication middleware, airspace management interface, cybersecurity framework, and analytics/archival modules.
- 4) Requires consistent access to external networks including telecommunications systems, meteorological services, and healthcare information networks.
- 5) Operates continuously, typically 24/7, to support real-time delivery requests and maintain supervision over MUAV activity.
- 6) Includes redundancy in power, communication, and processing units to ensure uninterrupted operations even during partial system degradation.

D. Attributes

- 1) Uses high-availability servers and networking equipment capable of real-time telemetry processing.
- 2) Uses redundant communication modems and antennas capable of automatically switching between cellular, satellite, and RF links.
- 3) Uses mission planning and optimization algorithms with the ability to consider communication coverage maps, weather forecasts, no-fly zones, and hospital time constraints.
- 4) Uses visualization dashboards that display MUAV locations, communication link status, and mission timelines.
- 5) Uses real-time alerting systems for communication loss, geofence violations, unexpected weather, or aircraft health anomalies.
- 6) Uses secure server clusters that store mission logs, communication transcripts, payload data, and regulatory records.
- 7) Uses operator-friendly HMIs enabling rapid decision-making and emergency command execution.

E. Capabilities

- 1) Be able to maintain continuous command-and-control communication with MUAVs throughout all phases of flight.
- 2) Be able to process and display MUAV telemetry including health metrics, location, communication link health, and payload condition.
- 3) Be able to plan and schedule MUAV missions according to hospital needs, mission priority, and available fleet capacity.
- 4) Be able to provide autonomous or semi-autonomous decision support, such as automatically rerouting aircraft

during communication degradation or weather hazards.

- 5) Be able to initiate emergency flight commands, including return-to-base, hover-and-hold, or divert landing.
- 6) Be able to detect communication anomalies and perform seamless failover between network layers.
- 7) Be able to interface with UTM/ATC systems to receive airspace advisories and update MUAV flight plans accordingly.
- 8) Be able to record and archive operational data, ensuring traceability, regulatory compliance, and system performance evaluation.
- 9) Be able to provide mission status updates to hospitals, clinics, and ground personnel via digital communication channels.

F. Performance Parameters

- 1) X percent uptime for command-and-control channels across all MUAV missions.
- 2) Command uplink latency under X ms under nominal conditions.
- 3) Telemetry refresh rate over X Hz for position data.
- 4) Automatic network switching must occur within X seconds without loss of command authority.
- 5) Intrusion detection must trigger alerts within X seconds.
- 6) Must remain operational in controlled ambient conditions typically between X°C and X°C with appropriate humidity regulation.
- 7) Mission logs and telemetry must be stored for over X years or regulatory equivalent.

G. Products

- 1) Improved medical logistics throughput.
- 2) Enhanced operational safety.
- 3) Optimized MUAV fleet performance.
- 4) Greater predictability in clinical workflows.
- 5) Regulatory compliance and audit readiness.

H. By-Products

- 1) High operational cost due to staffing, equipment, communication plans, and infrastructure.
- 2) Elevated energy consumption from servers, displays, climate control, and network equipment.
- 3) Lifecycle electronic waste from periodic hardware upgrades.
- 4) Increased dependence on digital infrastructure, making some manual processes obsolete or impractical.
- 5) Operational noise and electromagnetic emissions from communication equipment

I. Services

- 1) Provides mission control and supervision by ensuring all MUAV flights remain within safe operational parameters.
- 2) Maintains robust communication infrastructure by enabling continuous link with MUAVs and coordination among stakeholders.
- 3) Supports emergency medical logistics by coordinating time-critical deliveries with minimal delay.
- 4) Communicates with hospitals and Unmanned Traffic Management (UTC) authorities, ensuring both logistical alignment and legal compliance.
- 5) Archives operational data for auditing and improvement by supporting performance analytics and incident investigations.

J. Inputs

- Acceptable Inputs
 - 1) Valid operator commands issued through authorized consoles.
 - 2) Accurate telemetry streams from MUAVs.
 - 3) External data from approved sources.
 - 4) Communications signals within acceptable power, bandwidth, and latency thresholds.
 - 5) Stable electrical power inputs.
 - 6) Environmental conditions within operational bounds.
- Unacceptable Inputs
 - 1) Unauthorized or spoofed commands.
 - 2) Corrupted, stale, spoofed, or missing telemetry data.
 - 3) External data that is incompatible, outdated, or unverified.
 - 4) Communication signals below minimum thresholds.
 - 5) Unstable or insufficient power.
 - 6) Environmental exposures beyond tolerance.

K. Outputs

- Acceptable Outputs
 - 1) Timely and correct commands transmitted to MUAVs
 - 2) Accurate and usable telemetry visualizations.
 - 3) Secure communications to stakeholders.
 - 4) Warnings, alerts, and decision-support notifications.
 - 5) Stable execution of internal functions.
- Unacceptable Outputs
 - 1) Incorrect, delayed, or conflicting MUAV commands.
 - 2) Misrepresented telemetry displays or missing status updates.
 - 3) Failure to issue alerts during abnormal or emergency conditions.
 - 4) Security lapses.
 - 5) System silence or total loss of communication.

L. Stakeholders

- 1) Primary Operators: Personnel responsible for mission execution, telemetry monitoring, and decision-making. Their training, adherence to procedures, and response times directly influence system effectiveness.
- 2) Communication Engineers and Network Technicians: Ensure that communication modules, antennas, gateways, routing algorithms, and fallback links function as designed.
- 3) Mission Planners and Dispatch Coordinators: Convert hospital requests into executable missions, ensuring alignment between fleet capacity and medical urgency.
- 4) Maintenance Personnel: Support system uptime by maintaining racks, servers, power units, communication hardware, and environmental control equipment.
- 5) Healthcare Institutions: Provide mission demand, payload data, delivery windows, and chain-of-custody confirmations.
- 6) Telecommunication Providers: Offer communication infrastructure required for MUAV connectivity across ground and aerial environments.

- 7) System Designer / Manufacturer: Responsible for architecture design, software development, testing, and continuous improvement.
- 8) Aviation Regulators: Ensure legal compliance and safe integration with other airspace users.
- 9) Insurance and Risk Management Entities: Provide financial mechanisms and operational incentives to reduce risk and maintain safety standards.
- 10) Emergency Response Agencies: Enable coordinated multi-agency operations during medical emergencies or natural disasters.

M. Constraints

- 1) Communication coverage may be inconsistent in rural, mountainous, or dense urban regions.
- 2) CCC relies on external telecommunications infrastructure that may not be under its control.
- 3) BVLOS regulations impose communication reliability and visibility requirements.
- 4) Environmental constraints limit hardware durability.
- 5) System cost and staffing requirements restrict deployment scale.
- 6) Airspace requirements may impose delay or limit mission flexibility.

N. Threats

- 1) Communication signal jamming or interference from environmental or malicious sources.
- 2) Cyberattacks that target command channels or telemetry feeds.
- 3) Power infrastructure failures causing sudden loss of operations.
- 4) Hardware fatigue or software bugs that degrade performance over time.
- 5) Human error during mission planning or oversight.
- 6) Operational downturns that reduce funding for continuous operation.

O. Opportunities

- 1) Growing adoption of medical aerial logistics increases demand for CCC infrastructure.
- 2) Advances in communication technology (5G NTN, LEO satellites) increase resilience.
- 3) Increased digitization of hospitals integrates better with CCC data flows.
- 4) Development of standardized urban air corridors elevates demand for robust command centers.
- 5) Autonomous fleet management improvements reduce operator workload.
- 6) Emergency response agencies increasingly adopt UAV-based logistics.

III. System of Interest Concept of Operation

A. Context Diagram

The Command and Control Center (CCC) forms the core of the Medical Unmanned Aerial Vehicle (MUAV) System-of-Systems, integrating power and electrical components, facility infrastructure, computer hardware and software, the communication network, and the operator console. These internal elements work together to plan missions, monitor UAV status, and coordinate safe flight operations.

Surrounding the CCC are external systems that interact with it to support medical logistics. The delivery vehicle and its payload module exchange flight data, telemetry, and mission updates with the CCC, while UTM/ATC provides airspace coordination and routing approvals. Medical facilities and the medical transport system communicate needs, receive status updates, and link directly to UAV delivery operations. Logistics infrastructure, maintenance and charging

systems, and the broader network system also interface with the CCC to ensure vehicle readiness, data flow, and timely transport of medical supplies.

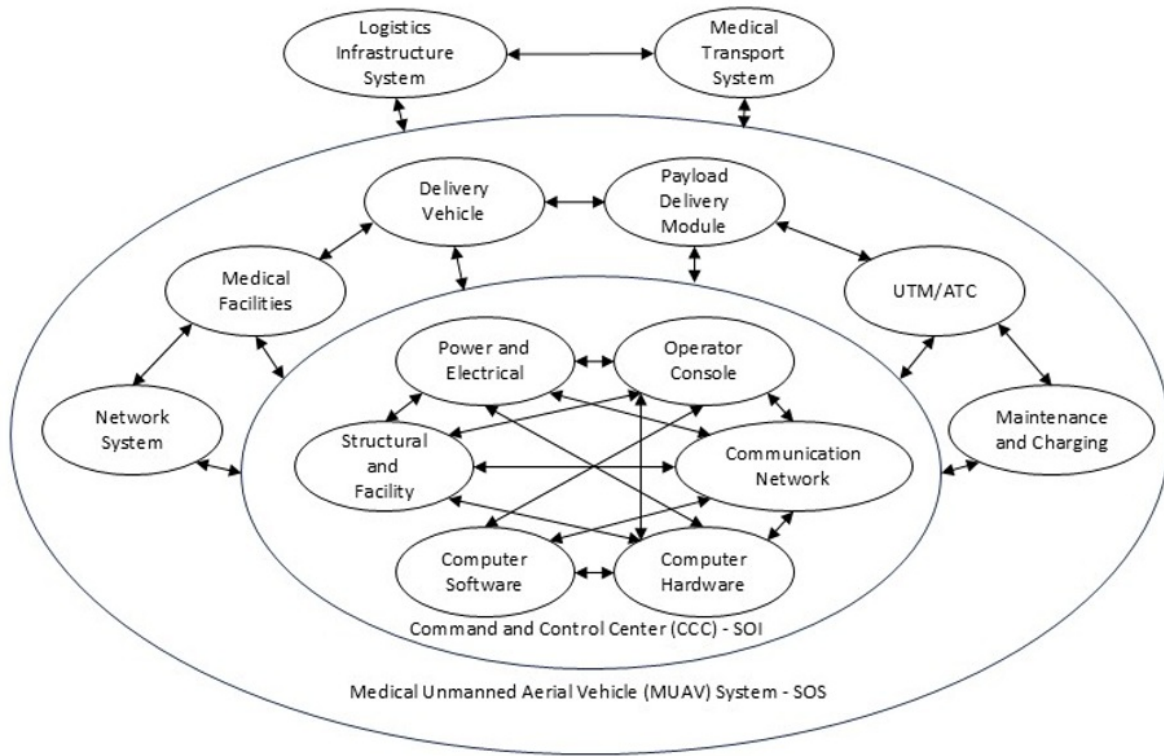


Fig. 2 Context Diagram

B. Concept of Operations

The Concept of Operations (ConOps) for the Command and Control Center (CCC) describes how the CCC orchestrates all activities required to execute medical UAV missions from initial request to final delivery. The process begins when a medical facility submits a mission request, which enters the CCC as operational input. The CCC first engages in mission planning (1.1), determining flight paths, delivery timing, required UAV assets, and coordination needs. This planning stage relies heavily on communication activities (1.2), where the CCC exchanges data and updates with the delivery vehicle, payload delivery module, network system, and other external stakeholders shown in the SV-1 diagram.

Once a preliminary mission plan is established, the CCC transitions to airspace coordination (1.3), sending airspace clearance requests to UTM/ATC and integrating their feedback to adjust routing and ensure regulatory compliance. With airspace secured, the CCC continually monitors (1.4) mission progress, tracking cargo status, UAV telemetry, and environmental conditions. This monitoring loop feeds back into the communication and planning functions, allowing the CCC to refine or adapt operations in real time.

Parallel to mission execution, the CCC also performs fleet management (1.5), using updates from the delivery vehicle and the maintenance and charging system to issue maintenance orders, track readiness levels, and ensure UAV availability for current and future missions. This fleet management function provides ongoing support to both mission planning and monitoring, ensuring that operational decisions are grounded in accurate, current fleet status.

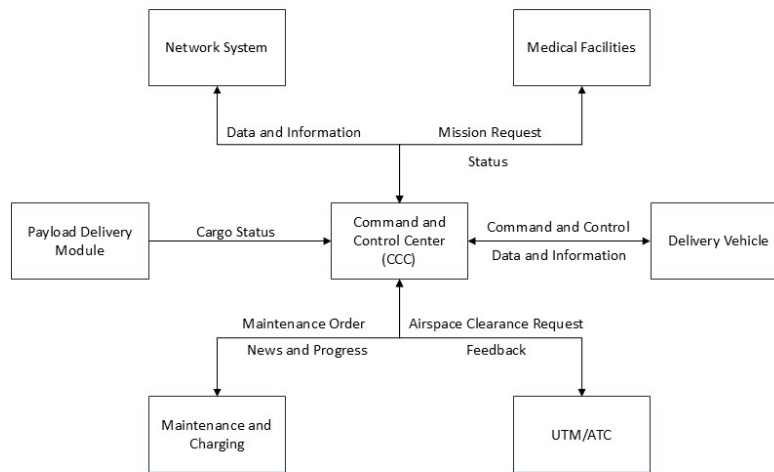


Fig. 3 SV-1 Diagram

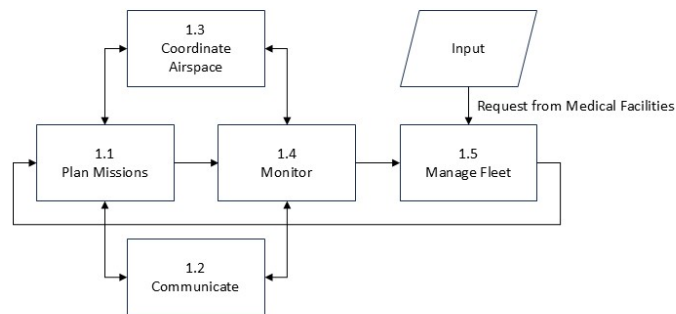


Fig. 4 Concept of Operations

IV. System Decomposition

A. Functional Decomposition

- **1.1 Plan Missions:** Encompasses all preparatory activities required to define, authorize, and prepare a mission for execution.
 - **1.1.1 Define Mission Requirements:** Formal specification of objectives, scope, and performance parameters.
 - **1.1.2 Conduct Route and Feasibility Analysis:** Assessment of operational constraints, potential routes, and technical viability.
 - **1.1.3 Approve Mission and Secure Authorization:** Obtaining formal clearance and necessary permissions from governing bodies.
 - **1.1.4 Verify Preflight Readiness:** Final confirmation that all personnel, assets, and systems are prepared for launch.
 - **1.1.5 Initiate Launch and Establish Supervision:** Execution of the launch sequence and transition to active mission oversight.
- **1.2 Communicate:** Responsible for establishing, maintaining, and securing reliable data and command links throughout the mission lifecycle.
 - **1.2.1 Establish Primary Communications:** Setting up the main operational communication channel.
 - **1.2.2 Maintain Backup Links:** Ensuring redundant communication pathways for operational resilience.
 - **1.2.3 Transmit Commands and Controls:** Sending operational instructions and real-time adjustments to the assets.
 - **1.2.4 Ensure Secure Network:** Implementing and verifying network security protocols to protect data integrity and command authority.
- **1.3 Coordinate Airspace:** Manages the interface with external airspace authorities to ensure regulatory compliance and operational safety.
 - **1.3.1 Submit Flight Plan to Airspace Authority:** Formal submission of the mission profile to regulatory bodies.
 - **1.3.2 Receive Airspace Clearance:** Obtaining official authorization to operate within the designated airspace.
 - **1.3.3 Monitor Airspace Compliance:** Continuous surveillance to ensure assets adhere to approved flight paths and restrictions.
- **1.4 Monitor:** Covers real-time surveillance, analysis, intervention, and reporting during active mission execution.
 - **1.4.1 Observe Real Time Status:** Continuous tracking of asset state and operational parameters.
 - **1.4.2 Analyze Telemetry Data:** Interpretation of streaming sensor and performance data for operational assessment.
 - **1.4.3 Display Actionable Information:** Presentation of analyzed data in a format suitable for rapid operator decision-making.
 - **1.4.4 Execute Operator Interventions:** Implementation of manual or automated adjustments to mission or asset behavior.
 - **1.4.5 Confirm Mission Progress:** Verification that execution is following the plan and objectives are being met.
- **1.5 Manage Fleet:** Focuses on strategic and tactical allocation and tracking of all MUAV assets.
 - **1.5.1 Schedule Missions:** Planning mission timing and sequencing based on fleet availability.
 - **1.5.2 Allocate MUAVs:** Assigning specific UAVs to missions according to capabilities and readiness.

- **1.5.3 Track All Active Missions:** Maintaining consolidated, up-to-date status of all ongoing missions and deployed assets.

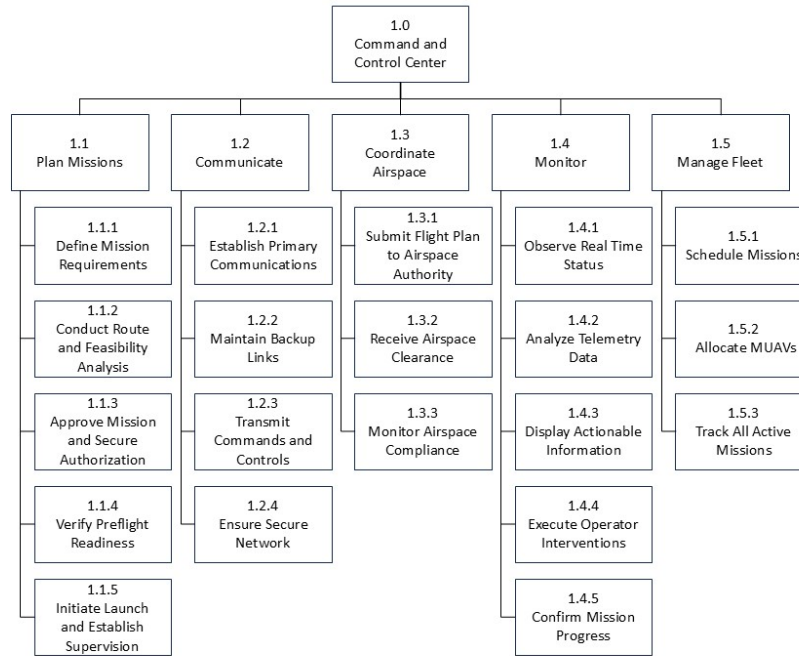


Fig. 5 Functional Decomposition

B. Functional Flows

The Command and Control Center (CCC) executes a seamless mission workflow through the integration of five primary functions, starting with the proactive Manage Fleet (1.5) function. This function accepts mission requests from external Medical Facilities and other inputs, using real-time MUAV Data to Schedule Missions and Allocate MUAVs efficiently, culminating in the creation of a definitive Mission package for the planning phase.

The Plan Missions (1.1) function then takes charge, initiating with Define Mission Requirements and Conduct Route and Feasibility Analysis. The mission proceeds to Approve Mission and Secure Authorization only after sending a Request to the Coordinate Airspace (1.3) function. Airspace coordination, which includes submitting the flight plan, receiving clearance, and monitoring compliance, provides critical feedback to planning and monitoring before the mission is launched. The planning phase concludes by verifying preflight readiness and initiating the launch, formally enacting the Transfer of MUAV Responsibility to the monitoring system.

As the flight commences, the Communicate (1.2) function maintains continuous operational integrity by establishing both Primary Communications and Backup Links. This function ensures a Secure Network for reliably transmitting mission Commands and receiving essential MUAV Data to and from all other primary functions.

The Monitor (1.4) function assumes continuous supervisory control, beginning with Observe Real Time Status and Analyze Telemetry Data. It synthesizes this information with Airspace Compliance data from the coordination function to Display Actionable Information for the operator. Crucially, the system supports Execute Operator Interventions via communicated Command signals when anomalies are detected. Upon completion, the Monitor function confirms mission progress, feeding the final MUAV Data back into the Manage Fleet (1.5) function to update the global fleet status and close the loop.

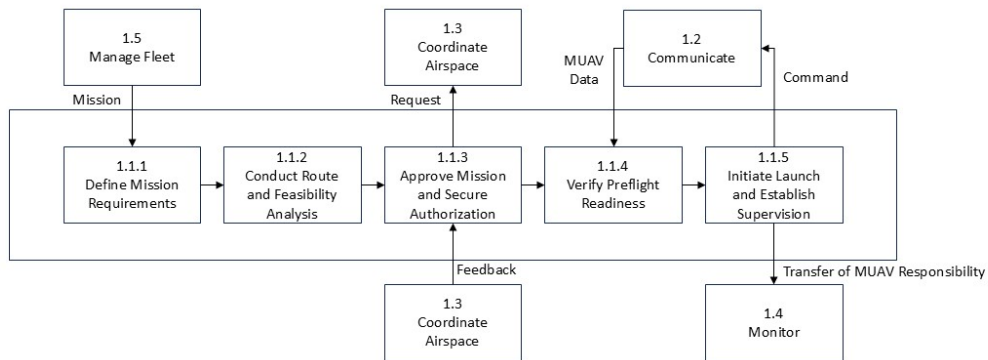


Fig. 6 Function 1.1 Plan Mission

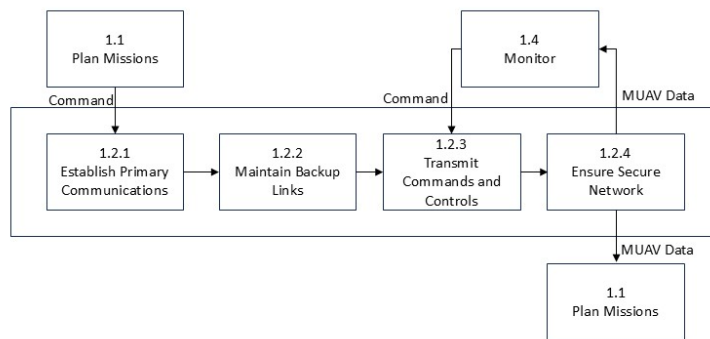


Fig. 7 Function 1.2 Communicate

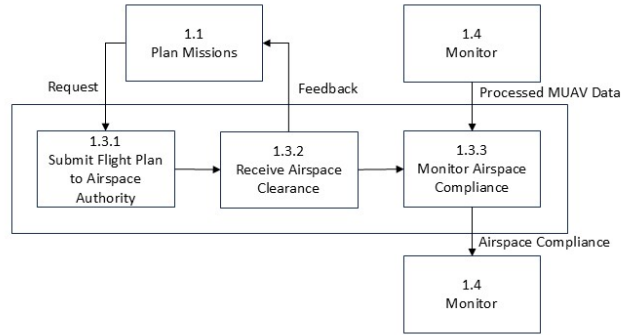


Fig. 8 Function 1.3 Coordinate Airspace

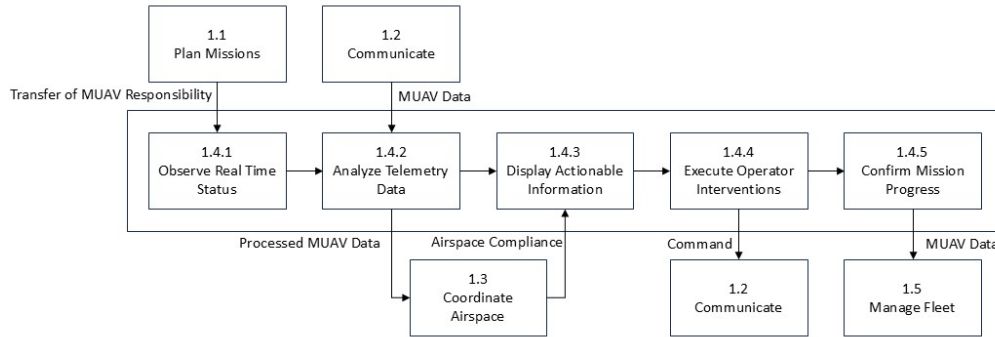


Fig. 9 Function 1.4 Monitor

C. Functional Timeline

The Command and Control Center (CCC) operates based on a precise temporal profile, with the Manage Fleet (F1.5) function serving as the operational core, running continuously from minute 0 through minute 28 or longer. A

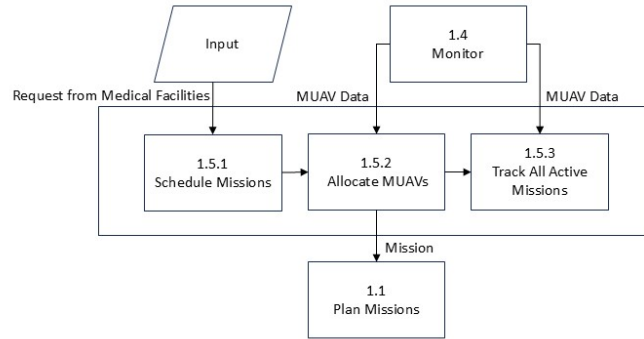


Fig. 10 Function 1.5 Manage Fleet

mission is received at 5 minutes, which simultaneously activates the intensive Plan Missions (F1.1) function. F1.1 runs for a compressed period from 5 minutes to 9 minutes, focusing on route feasibility, securing airspace authorization, and preflight verification. The MUAV departs from base at 6 minutes, a moment that triggers the primary operational functions: Communicate (F1.2) and Coordinate Airspace (F1.3). These two functions run concurrently from 6 minutes until 24 minutes, responsible for maintaining the secure command link and ensuring the MUAV adheres to all regulatory airspace constraints throughout the flight. The Monitor (F1.4) function assumes in-flight supervisory control starting at 7 minutes and remains active until 24 minutes, encompassing all key mission milestones, including arrival at pick-up at 12 minutes, departure at 13 minutes, arrival at drop-off at 16 minutes, departure at 17 minutes, and the final arrival at base at 24 minutes.

Time (min)	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Event						Mission Received	Depart from Base						Arrive at Pick-up	Depart Pick-up	Arrive at Drop-off	Depart Drop-off							Arrive at Base						
F1.1 Plan Missions																													
F1.2 Communicate																													
F1.3 Coordinate Airspace																													
F1.4 Monitor																													
F1.5 Manage Fleet																													

Fig. 11 Functional Timeline

D. Structural Decomposition

The system is built upon a resilient Structural and Facility (1.1) foundation that provides the Structural Enclosure and Environmental Control Unit, backed by a robust Power and Electrical (1.2) system featuring Battery Units and a Backup Generator for continuous, uninterrupted operation. The system's intelligence and primary function are defined by the convergence of the Communication Network (1.3)—which includes Antennas and Security and Encryption hardware—and the Computer Hardware (1.4), which utilizes Processing Servers and specialized Telemetry Hardware to ingest and process mission data. This processed data is then utilized by the Computer Software (1.5), which contains the operational logic for Mission Planning, Monitoring, and the Airspace Interface required for regulatory compliance. All this complexity culminates at the Operator Console (1.6), which is the system's human-machine interface, composed of Workstations, Display Panels, and Input and Output Devices used to efficiently coordinate the overall fleet.

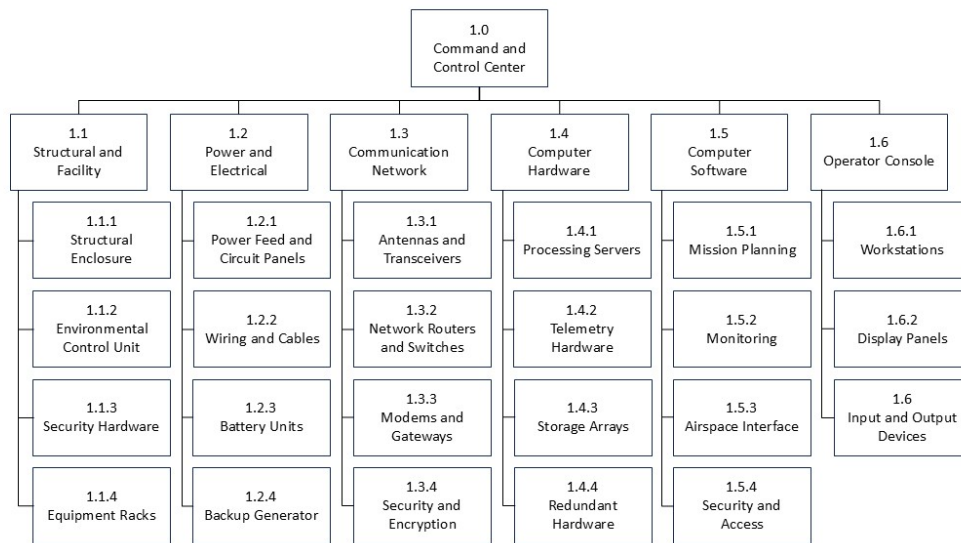


Fig. 12 Structural Decomposition

E. Interfaces

The N2 diagram reveals a distinct hierarchical integration pattern where the functional products are heavily cross-linked, demonstrating the complexity of the Command and Control Center (CCC) as a system of systems. The highest density of interfaces is concentrated among the Communication Network (1.3), Computer Hardware (1.4), and Computer Software (1.5), establishing them as the core processing and logic engine. Specifically, the Computer Hardware—including Processing Servers (1.4.1) and Telemetry Hardware (1.4.2)—functions as the central hub, acting as an intermediary that interfaces with nearly every component in the Communication Network to receive data and with every module in the Computer Software suite (1.5.x) to execute logic. This software logic, encompassing Mission Planning and Monitoring, then directly drives the Operator Console (1.6), showing a strong Human-Machine Interface (HMI) dependency on the processed functional output. In contrast, supporting infrastructure like the Structural (1.1) and Power/Electrical (1.2) subsystems exhibit minimal direct connections to the complex operational software (1.5), confirming their role as isolated utility providers that only interface with the base hardware for physical stability and

power supply.//

Product Interfaces N ²	1.1 Structural and Facility	1.1.1 Structural Enclosure	1.1.2 Environmental Control Unit	1.1.3 Security Hardware	1.1.4 Equipment Racks	1.2 Power and Electrical	1.2.1 Power Feed and Circuit Panels	1.2.2 Wiring and Cables	1.2.3 Battery Units	1.2.4 Backup Generator	1.3 Communication Network	1.3.1 Antennas and Transceivers	1.3.2 Network Routers and Switches	1.3.3 Modems and Gateways	1.3.4 Security and Encryption	1.4 Computer Hardware	1.4.1 Processing Servers	1.4.2 Telemetry Hardware	1.4.3 Storage Arrays	1.4.4 Redundant Hardware	1.5 Computer Software	1.5.1 Mission Planning	1.5.2 Monitoring	1.5.3 Airspace Interface	1.5.4 Security and Access	1.6 Operator Console	1.6.1 Workstations	1.6.2 Display Panels	1.6.3 Input and Output Devices
	1.1 Structural and Facility																												
	1.1.1 Structural Enclosure		x	x	x		x	x	x	x		x	x	x			x	x	x	x							x	x	x
	1.1.2 Environmental Control Unit						x	x	x	x							x	x	x	x							x	x	x
	1.1.3 Security Hardware						x	x	x	x		x	x	x			x		x						x		x	x	x
	1.1.4 Equipment Racks						x	x				x	x	x			x	x	x	x							x	x	x
	1.2 Power and Electrical																												
	1.2.1 Power Feed and Circuit Panels							x	x	x			x	x	x	x		x	x	x	x						x	x	x
	1.2.2 Wiring and Cables								x	x			x	x	x	x		x	x	x	x						x	x	x
	1.2.3 Battery Units										x		x	x	x	x		x	x	x	x						x	x	x
	1.2.4 Backup Generator												x	x	x	x		x	x	x	x						x	x	x
	1.3 Communication Network																												
	1.3.1 Antennas and Transceivers													x	x	x		x	x	x	x						x	x	x
	1.3.2 Network Routers and Switches														x	x		x	x	x	x						x	x	x
	1.3.3 Modems and Gateways															x		x	x	x	x						x	x	x
	1.3.4 Security and Encryption																	x	x	x	x						x	x	x
	1.4 Computer Hardware																												
1.4.1 Processing Servers																		x	x	x							x	x	x
1.4.2 Telemetry Hardware																			x	x							x	x	x
1.4.3 Storage Arrays																				x							x	x	x
1.4.4 Redundant Hardware																											x	x	x
1.5 Computer Software																													
1.5.1 Mission Planning																											x	x	x
1.5.2 Monitoring																											x	x	x
1.5.3 Airspace Interface																											x	x	x
1.5.4 Security and Access																													
1.6 Operator Console																													
1.6.1 Workstations																												x	x
1.6.2 Display Panels																													x
1.6.3 Input and Output Devices																													

Fig. 13 N2 Diagram

V. System Requirements

Sequence	Product	Function	Requirement	Rationale	Type	Interface
----------	---------	----------	-------------	-----------	------	-----------

1	1.5, 1.6	1.1	The CCC shall allow operators to define mission requirements, including destination, payload type, delivery priority, and environmental constraints, through a structured interface within XXX seconds of user request.	Clear requirement definition ensures the MUAV receives complete and accurate mission parameters, enabling correct flight planning and reducing operator error.	FNC	Internal
2	1.5.1, 1.4.1	1.1.2	The CCC shall automatically compute a feasible MUAV route using terrain, weather, and regulatory data within XXX seconds, ensuring that the MUAV has sufficient range and safe operating conditions.	Fast and accurate route analysis reduces mission delays and supports timely medical delivery.	PER	Internal
3	1.3	1.1.3	The CCC shall submit mission flight plans to the airspace authority and receive authorization before launch, ensuring compliance with FAA/UTM BVLOS operational rules.	Legal authorization prevents airspace violations and ensures safe MUAV integration with other aircraft.	REG	External
4	1.5.2	1.1.4	The CCC shall verify MUAV readiness, including battery level, communication link, payload status, and system health, within XXX seconds prior to launch.	Preflight checks reduce the likelihood of mid-flight failures and increase mission reliability.	FNC	Internal
5	1.3	1.1.5	The CCC shall initiate MUAV launch only after all verifications complete, and shall establish continuous supervision within XXX seconds of takeoff.	Ensures the MUAV starts the mission safely and under real-time oversight.	FNC	Internal
6	1.3	1.2.1	The CCC shall establish a primary MUAV communication link with a minimum throughput of XXX Mbps and latency no greater than XXX ms.	Low-latency, reliable communication is necessary for transmitting commands and receiving telemetry in real time.	FNC	External

7	1.3.3	1.2.2	The CCC shall maintain at least one redundant communication link capable of assuming control within XXX milliseconds of primary link failure.	Redundancy prevents catastrophic loss of control due to communication dropouts.	CST	External
8	1.3.1	1.2.3	The CCC shall transmit command uplinks to MUAVs at a frequency of at least XXX Hz, ensuring prompt control response.	Consistent uplinks maintain safe vehicle operation and timely updates.	FNC	External
9	1.3.4	1.2.4	The CCC shall encrypt all communications with at least XXX-bit encryption and implement authentication protocols compliant with XXX cybersecurity standard.	Prevents unauthorized control takeovers and protects payload and mission data.	CST	External
10	1.5.3	1.3.1	The CCC shall automatically submit a flight plan to the airspace authority within XXX seconds after route approval.	Ensures timely integration into shared airspace environments.	FNC	External
11	1.5.3	1.3.2	The CCC shall receive and process airspace clearance updates within XXX seconds of issuance by the airspace authority.	Quick processing prevents delays and airspace conflicts.	FNC	External
12	1.5.3	1.3.3	The CCC shall continuously monitor MUAV compliance with assigned airspace corridors and geofences with a tracking refresh rate of XXX Hz.	Supports safety, coordination, and regulatory adherence.	CST	External
13	1.6.2	1.4.1	The CCC shall maintain real-time visibility of MUAV location, altitude, speed, and payload status with data updated every XXX ms.	Enables operators to track mission status and respond rapidly to anomalies.	FNC	Internal

14	1.4.2	1.4.2	The CCC shall process incoming telemetry and identify anomalies within XXX seconds using automated analysis algorithms.	Rapid anomaly detection enhances mission safety and reduces operator workload.	PER	Internal
15	1.6.2	1.4.3	The CCC shall display prioritized, actionable alerts with a maximum operator recognition time of XXX seconds.	Ensures that operators can quickly understand and address critical events.	FNC	Internal
16	1.6.3	1.4.4	The CCC shall allow operators to issue manual control inputs that take effect on the MUAV within XXX milliseconds.	Rapid manual control is required for emergencies and non-nominal conditions.	FNC	Internal
17	1.6.1	1.4.5	The CCC shall record mission completion and confirm delivery status within XXX seconds of MUAV arrival.	Provides accountability, traceability, and supports post-mission analysis.	FNC	Internal
18	1.5.1	1.5.1	The CCC shall schedule missions to avoid vehicle conflicts, with schedule resolution accuracy of XXX minutes.	Effective scheduling optimizes fleet utilization and mission timeliness.	FNC	Internal
19	1.5.1	1.5.2	The CCC shall select an MUAV for each mission using availability, battery level, and payload compatibility criteria computed within XXX seconds.	Optimal asset assignment improves delivery reliability.	FNC	Internal
20	1.4.1	1.5.3	The CCC shall track the status and location of all active MUAVs simultaneously, updating global fleet status every XXX ms.	Fleet-wide awareness prevents mission interference and enhances safety.	FNC	Internal
21	1.3.1, 1.5.2	1.2.3	The MUAV shall transmit telemetry to the CCC at a minimum rate of XXX packets/second with end-to-end latency $\leq$ XXX ms.	Supports real-time monitoring and decision-making.	IR	External

22	1.3	1.2.3	The CCC shall provide control commands to the MUAV with a maximum uplink delay of XXX ms.	Timely uplinks ensure safe dynamic control of the aircraft.	IR	External
23	1.3.4	1.4.4	The CCC shall support a secure emergency override channel that remains functional during primary link failure and activates within XXX ms.	Allows operators to maintain emergency control while preventing unauthorized access.	IR/CST	External
24	1.5.3	1.3.1	The CCC shall transmit flight plans to the airspace authority using the UTM-compliant protocol and confirm receipt within XXX seconds.	Ensures regulatory compliance and timely mission approval.	REG/IR	External
25	1.5.3	1.3.2	The CCC shall receive clearance updates with a maximum processing delay of XXX seconds.	Quick updates prevent unauthorized flight operations.	REG/IR	External
26	1.6.1	nan	The CCC shall receive mission requests from healthcare facilities and process them within XXX seconds.	Fast request intake supports timely medical supply delivery.	IR	External
27	1.4.1	1.4.1–1.4.3	The CCC shall verify payload temperature, lock status, and identity with telemetry updates at least every XXX seconds.	Verifies chain-of-custody and ensures medical payload integrity.	IR/CST	External
28	1.3, 1.5	1.2.3	Communication hardware shall deliver data to software components with processing latency not exceeding XXX ms.	Supports real-time display and decision-making.	IR	Internal
29	1.4, 1.6	1.4.1	All processed mission data shall be delivered to operator workstations with an update frequency of XXX Hz.	Ensures operators receive timely and accurate information.	IR	Internal

30	1.2	1.2.2	All CCC subsystems shall receive uninterrupted power with total downtime not exceeding XXX minutes/year and automatic failover switching within XXX ms.	CCC uptime is critical for mission safety and continuous MUAV operations.	CST/IR	Internal
----	-----	-------	---	---	--------	----------
